

NULLROOT

root@nullroot.dev | Dar es Salaam, Tanzania (UTC+3) | github.com/nullroot | hackerone.com/nullroot

Computer engineering student · independent security researcher · MQL5 algo developer · full-stack web · embedded

PROFESSIONAL SUMMARY

Final-year computer engineering student working the overlap of offensive security, automation and hardware. Web-first vulnerability research against bug-bounty programs, with reports written for triagers — repro steps first, drama never. Builds Expert Advisors with real risk models, full-stack web tooling in React/TypeScript/Next.js, and embedded fleets on ESP32 and Raspberry Pi. The through line: break it on purpose, then write it up.

SELECTED WORK & RESEARCH

Independent Security Researcher (Bug Bounty)

Self-directed | 2023 - Present | Remote

- Discovered and reported authorization gaps (IDOR), injection and business-logic flaws across web and API targets; first valid report triaged and resolved within one business week.
- Recon automation in Python: weekly attack-surface snapshots with diff alerts on new subdomains, JS bundles and API routes.
- Custom Nuclei templates for edge-case classes; Burp Suite as the daily driver, extensions configured for repeater-heavy workflows.

Algorithmic Trading Developer (MQL5)

Self-directed | 2024 - Present | Remote

- Shipped MetaTrader 5 Expert Advisors with volatility-scaled sizing, session filters and hard drawdown breakers; four retired prototypes before one survivor passed a 90-day forward test on demo.
- Backtesting pipeline in Python (pandas, vectorbt) over years of tick data; strategy changes gated by out-of-sample results, not screenshots.

Embedded & Full-Stack Developer

Freelance / academic projects | 2022 - Present | Dar es Salaam

- POTHOLE_VISION: Raspberry Pi 4 + YOLO road-defect detector, 91.3% mAP@0.5 — defended as a final-year project and shipped as a public writeup.
- MQTT sensor fleet of 12 ESP32 nodes deployed for a campus lab; QoS tuning, OTA updates and a monitoring dashboard that stayed boring on purpose.
- Full-stack delivery in Next.js/TypeScript/Prisma, including this portfolio: a 14-cabinet browser arcade, trophy engine and a site-wide CTF, all client-side with zero console errors.

EDUCATION

B.Sc. Computer Engineering

University of Dar es Salaam | final year, expected 2026

Focus: digital systems, networking and signal processing. Coursework on autopilot; the real curriculum runs on a Pi cluster.

CERTIFICATIONS

- eJPT — INE Security, 2024
- CompTIA Security+ — 2024
- CCNA — Cisco NetAcad, 2023
- HackTheBox CDA (Web) — 2025
- OSCP — in progress, 2026
- Practical JS Reversing — in progress, 2026

TECHNICAL SKILLS

Offensive security: Burp Suite, Nmap, Nuclei, ffuf, httpx/dnsx, Python recon pipelines, IDOR/authz, XSS, API testing

Trading & data: MQL5, MetaTrader 5, pandas, vectorbt, backtest design, risk models

Web: TypeScript, React, Next.js, Tailwind, Prisma, WebSockets, Playwright/E2E

Embedded & tooling: C, Arduino, ESP32, Raspberry Pi, MQTT, Git, Linux (Arch daily driver), DSP fundamentals

Verification: everything above runs live at nullroot.dev — arcade, writeups, telemetry. References on request.